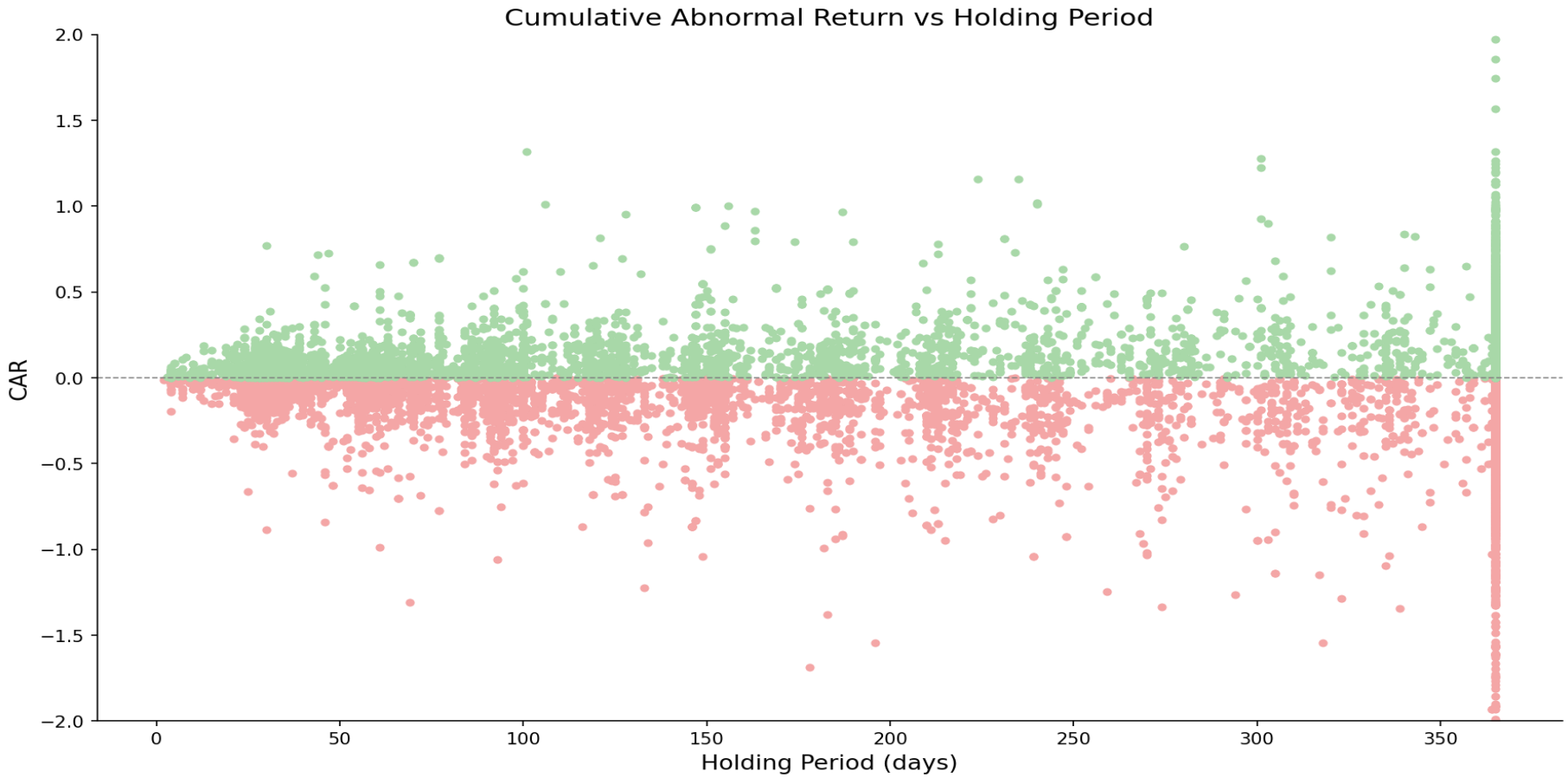
Congressional Stock Analysis

Do Congress Members Beat the Market?

Machine Learning on STOCK Act Disclosures · March 2026

Most Congressional Purchases Underperform the Market

Blindly copying congressional trades based on public disclosure dates **does not generate alpha**, the average politician underperforms their own risk-adjusted benchmark, making indiscriminate trade replication an inefficient strategy



15k
Total modeled trades

-0,5%
Median CAR

48.5%
Win rate

Holding period is calculated based on disclosed buy and sell date

How We Measure Performance: Cumulative Abnormal Return (CAR)

Cumulative Abnormal Return (CAR)	
Positive CAR (+) Compensated MORE for risk than CAPM predicts	Negative CAR (-) Compensated LESS, paid for risk that did not materialize

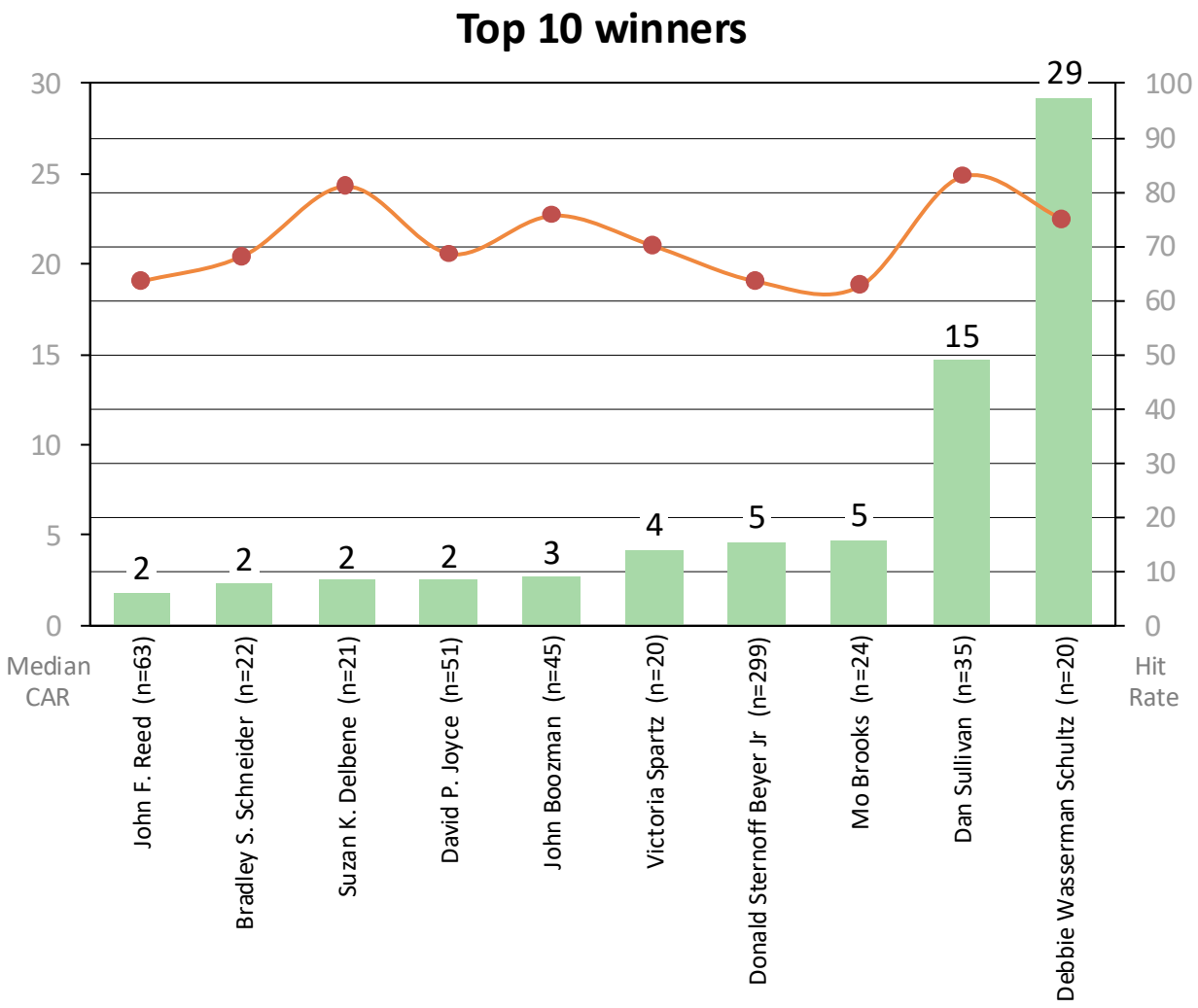
For each day, calculate the excess return relative to the stock's expected risk-adjusted return, then accumulate those daily excesses over the window.

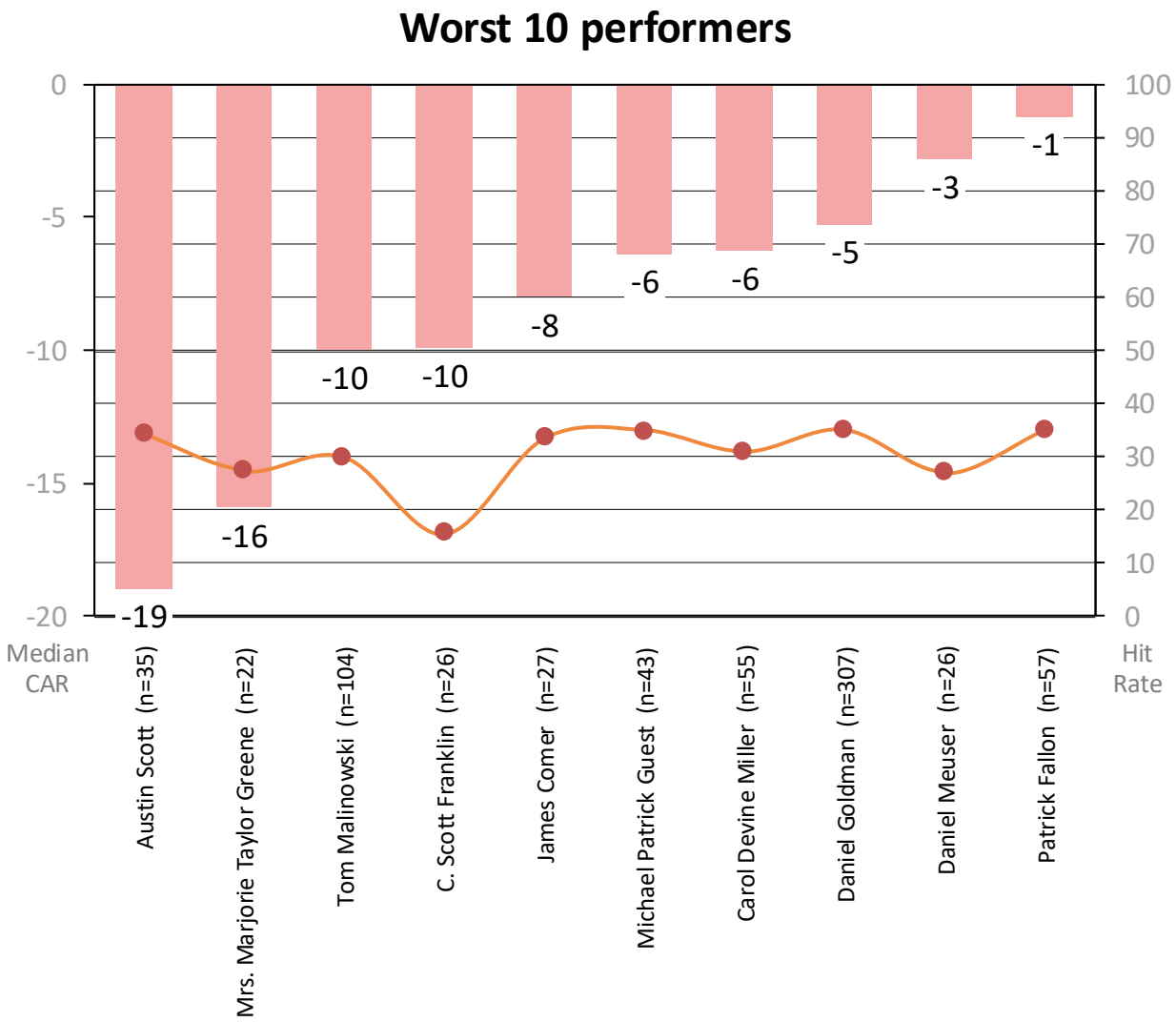
Excess return
actual log return –
CAPM-expected log return

CAPM
E[R] = R, + B x (Rm - R,)
Beta (B) estimated from 756 trading days of close prices (min 60 days).

Window
Measured buy-to-sell (if <12 months) or a fixed 12-month window.
The clock starts the day AFTER the public filing date

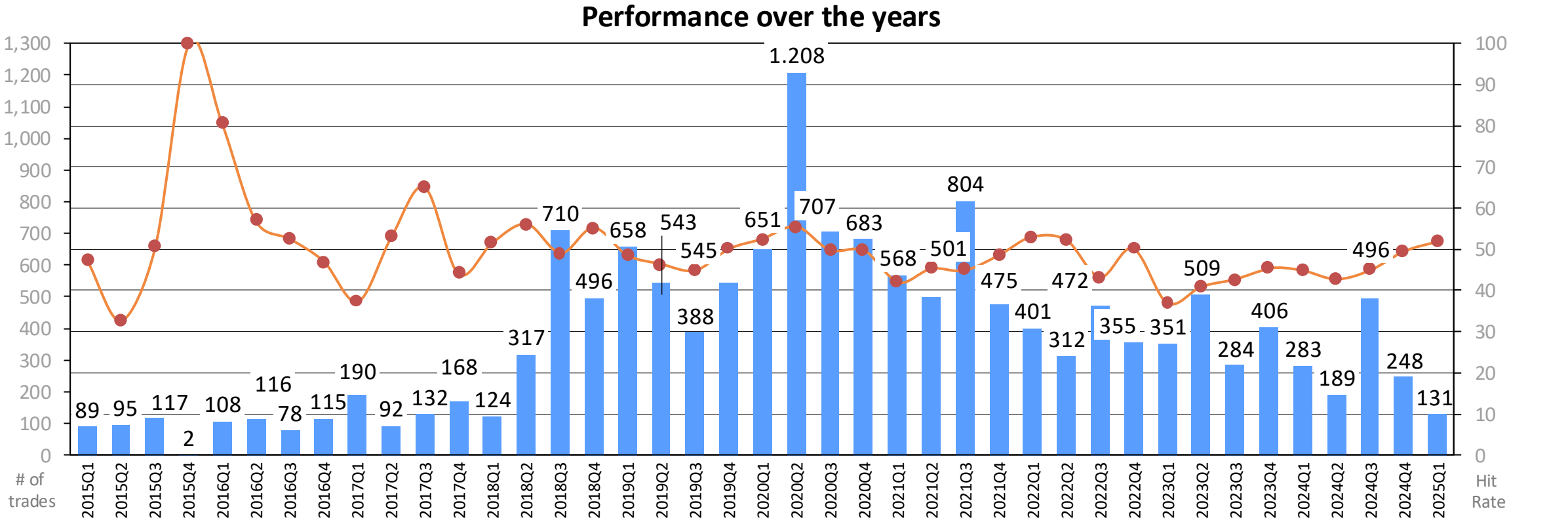
Who Consistently Beats — and Loses to — the Market?





- Based on 101 politicians with ≥ 20 disclosed purchases
- evaluated based on CAR performance, amount of $ traded was not considered

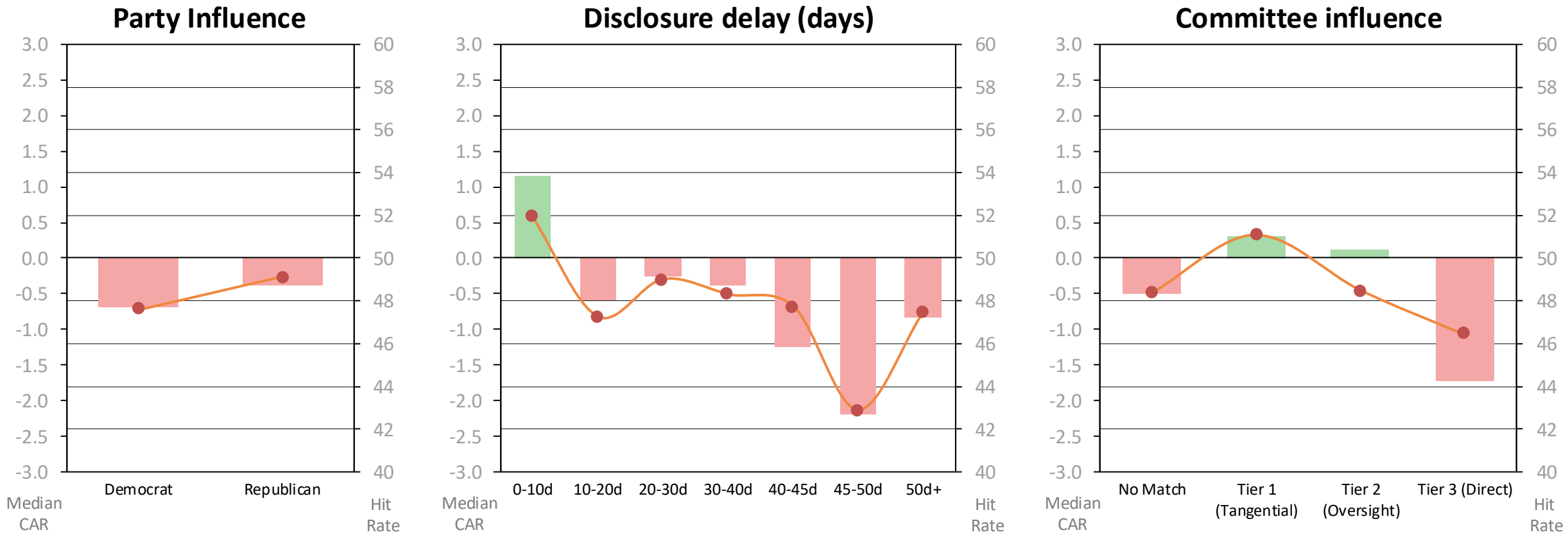
Performance Over Time: Macro Events Drive Positive Rates



- 2020 Q2–Q3: surge in positive rate — COVID recovery bounce lifted most purchases
- 2022: sharp drop aligned with rate-hike drawdown across the market
- Volume spikes suggest increased trading activity around election cycles
- Positive rate rarely exceeds 60% — even in the best macro environments

# of trades Hit Rate

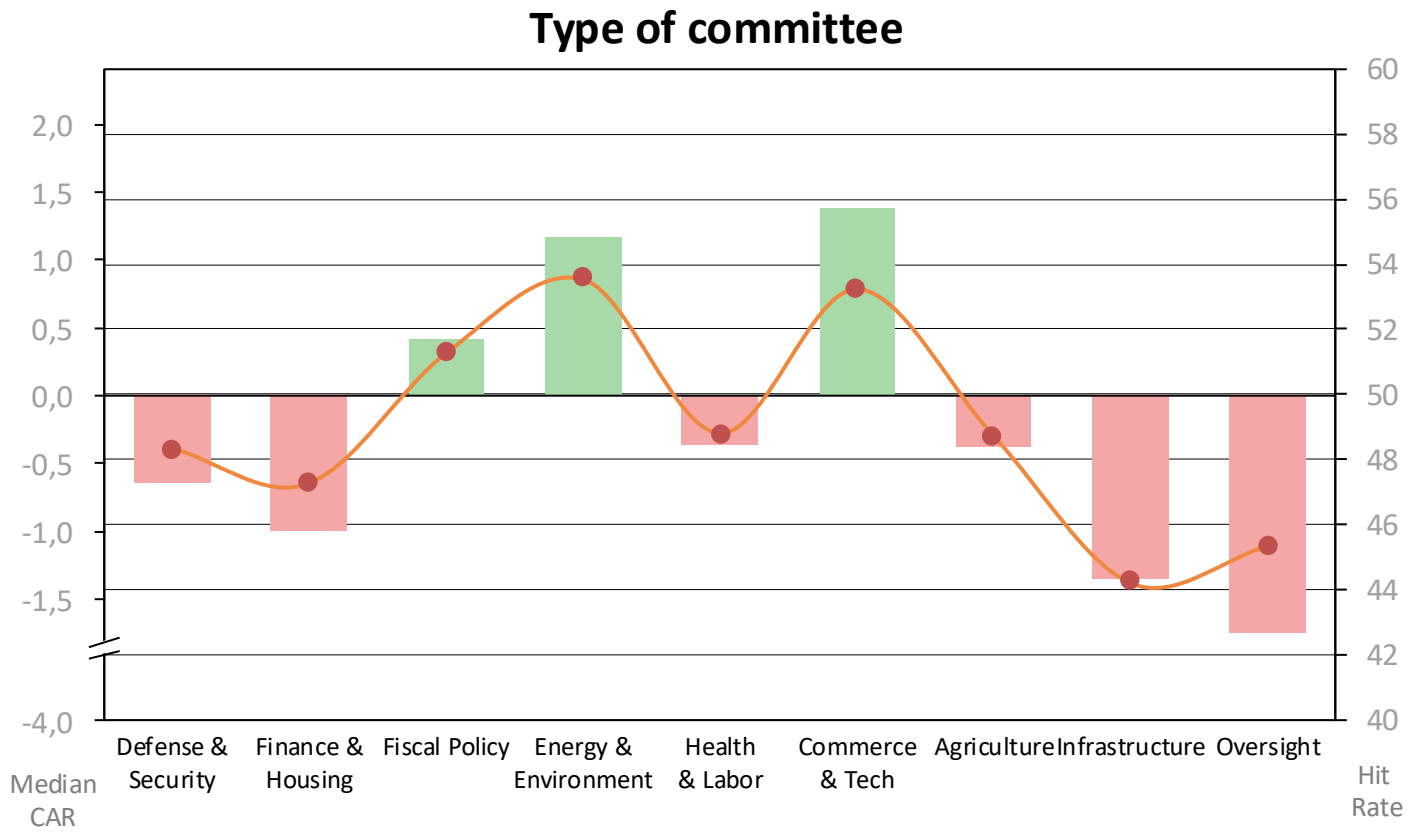
EDA: Median CAR Across Key Structural Variables

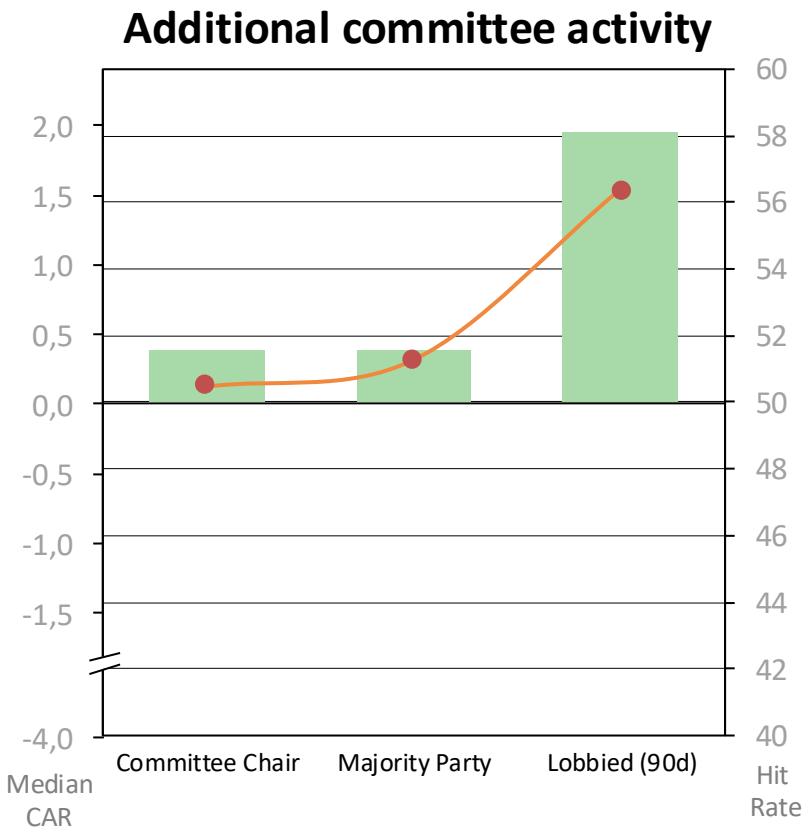


- Disclosure delay capped at 45 days by STOCK Act; 50d+ entries reflect late filers

Median CAR — Hit Rate

Committee Membership Shows Selective Advantage



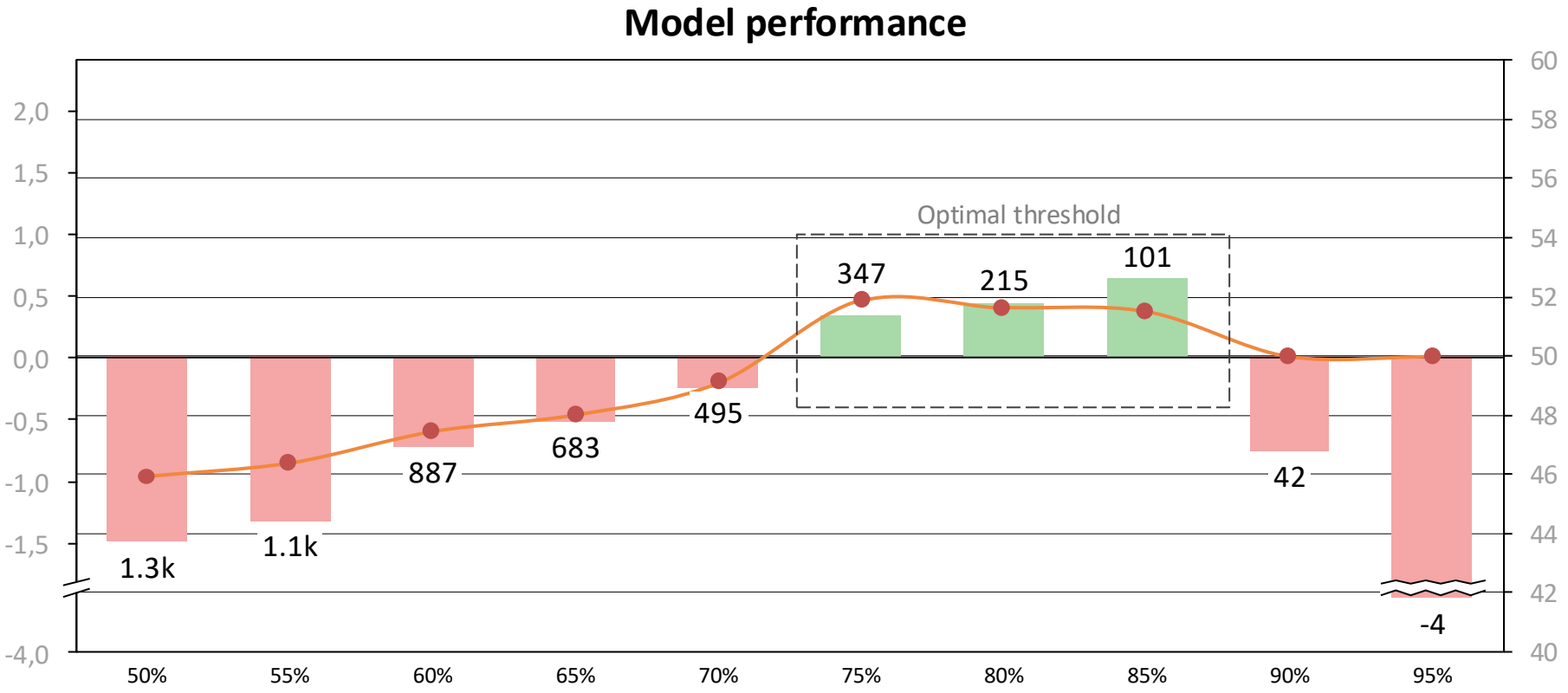


- Orange = flag=1 (on committee / in role / company lobbied) · Blue = flag=0
- All 9 committee categories shown — previously only 5 were displayed
- Lobbied (90d): company had active lobbying disclosure in 90 days before filing date

Median Car — Hit Rate

Can an AI Model do better?

A machine learning filter **recovers a subset of trades with positive risk-adjusted returns**, suggesting that **informed selection** not blind replication is where the edge lies



2
Years of prediction data

Median Car

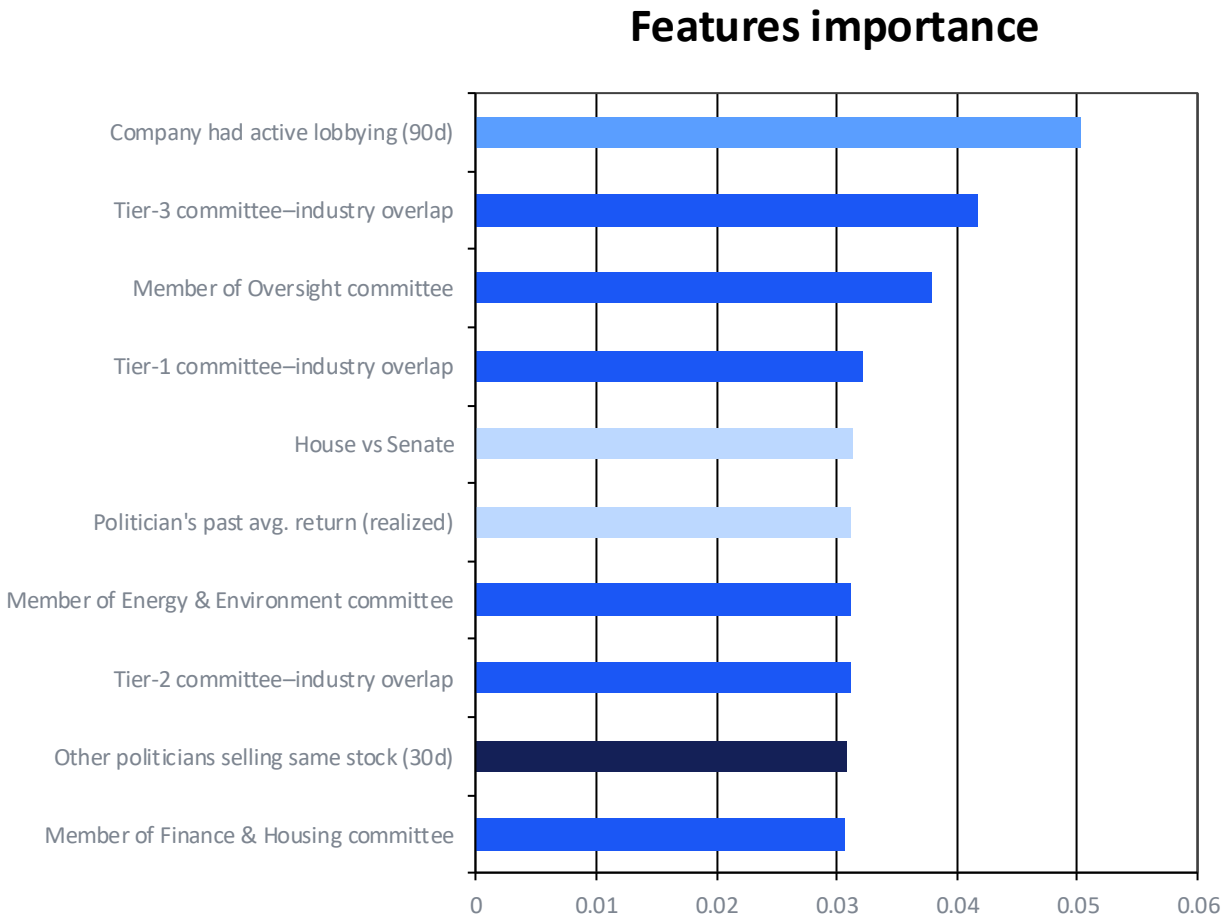
+0,5%	-1,95%
Optimal threshold	All trades

Hit Rate

51,5%	44,5%
Optimal threshold	All trades

Optimal strategy:
The model's value lies not in predicting every trade, but in knowing which ones to ignore, filtering congressional purchases down to the highest-confidence signals flips a losing strategy into a marginally profitable one

The Model: What Drives the Predictions?



Politician characteristics

Past politician performance

Lobbying activity

Intersections between traded companies and active lobbying

Committee Jurisdiction

Committee–industry jurisdictional overlap

Stock Characteristics

Information related to the stock

Trade Signals

Specific trade information

Data sources

1: https://api.quiverquant.com/beta/bulk/congresstrading
2: Yahoo Finance
3: https://github.com/unitedstates/congress-legislators
4: https://api.quiverquant.com/beta/historical/lobbying

Conclusions & Caveats

Key Findings

- **Copying congressional trades blindly destroys value** — across the full test set, the median trade delivers -1.95% CAR with only a 44.6% win rate, meaning the average politician earns less than what their stocks' own systematic risk predicts

- **Lobbying activity and committee jurisdiction are the strongest structural signals** — whether a company was actively lobbying in the 90 days before filing is the single most important model feature, followed by committee oversight and direct industry-jurisdiction overlap

- **Selective filtering recovers the edge** — a machine learning model identifies a subset of trades with positive risk-adjusted returns, suggesting informed selection is where the alpha lies

Caveats

- **Single macro regime** — the test set covers Dec 2022 – Jan 2025 only; performance across different market environments (e.g. bull runs, rate cycles) is not validated out-of-sample

- **Execution constraints are not modelled** — no transaction costs, market impact, or liquidity limits are accounted for; real-world returns will be lower, particularly for less liquid tickers

- **Disclosed trades only** — STOCK Act compliance is imperfect; undisclosed or late-filed activity is not captured, meaning the dataset may systematically miss the most sensitive trades

Potential Data Enrichment (Lobbying Focus)

- **Lobbying depth** — currently only a binary 90-day presence flag; adding spend amount, trend direction, and which specific bills overlap with the politician's committee jurisdiction would sharpen the signal

- **Proximity signal** — days between the lobbying registration date and the trade's filing date; tighter proximity likely indicates more actionable information access

- **Revolving door** — identifying lobbyists who previously worked for the same politician or committee would flag trades with unusually direct information channels

Thank you

Methodology: How Key Signals Were Constructed

1 — Ticker ↔ Committee Matching

- Committee membership from timestamped YAML snapshots (unitedstates/congress-legislators on GitHub)
- For each trade, the snapshot closest in time to the Filed date is used
- Ensures membership reflects the politician's actual committees at trade time
- Committee codes truncated to 4 characters to normalize across snapshot formats
- Result: binary committee-category flags per trade (9 categories total)

2 — 3-Tier Committee–Industry Match

- Manual mapping in config/commette_industry_map.yaml: committee → sector → industry
- Tier 3 (Strong): committee has DIRECT JURISDICTION — controls budgets, contracts, or legislation
- Tier 2 (Medium): committee has OVERSIGHT — sets general rules, taxes, or compliance standards
- Tier 1 (Tangential): committee has INDIRECT impact — general legislative influence only
- Result: Industry match 1/2/3 binary flags per trade (1 = politician's committee matches stock's industry at that tier)

3 — Lobbying Signal

- Source: QuiverQuant lobbying disclosures API, downloaded per traded ticker
- lobbied_any_90d = 1 if the company had any active lobbying disclosure in 90 days before trade's Filed date
- Currently binary — lobbying spend amount and issue specificity are future enrichment candidates

APPENDIX

Dataset Overview

39,334	19,096	15,117	268
Total Records	Purchases	Modelled	Politicians

Metric	Value
Date range	2012-09-13 – 2026-01-30
Transaction types	Purchase / Sale / Sale (Full) / Sale (Partial) / Exchange
Unique tickers (purchases)	2,209
Parties	D / R / I
Chambers	Representatives / Senate
Target variable	realized_car_hybrid > 0 (binary)
CAR windows available	1m / 3m / 6m / 9m / 12m + realized
Train / test split	80% / 20% chronological